

Blind Hole Thread Depth Charts

Co-op Tool Mfg Preferences

Use these charts to model full thread depth and tap-drill depth to the full-diameter shoulder. Metric depths are shown in inches and millimeters; UNC depths are in inches. Metric sizes use coarse pitch.

Inch full thread depth = $1.5 \times \text{diameter (in)} + .125''$

Metric full thread depth = $1.5 \times \text{diameter (mm)} + 3.25 \text{ mm}$

Extra drill depth = greater of $.125''$ for inch threads or 3.25 mm for metric threads, or $2.5 \times \text{pitch}$

Tap Drill Depth = full thread depth + extra drill depth

Metric Coarse Threads M3 to M24

Thread callouts give nominal diameter $\times$ pitch in millimeters.

Thread size	Full thread depth (in)	Full thread depth (mm)	Tap Drill Depth (in)	Tap Drill Depth (mm)
M3 $\times$.5	0.31	7.8	0.44	11.1
M3.5 $\times$.6	0.34	8.5	0.47	11.8
M4 $\times$.7	0.37	9.3	0.50	12.6
M5 $\times$.8	0.43	10.8	0.56	14.1
M6 $\times$ 1	0.49	12.3	0.62	15.6
M7 $\times$ 1	0.55	13.8	0.68	17.1
M8 $\times$ 1.25	0.61	15.3	0.74	18.6
M10 $\times$ 1.5	0.72	18.3	0.87	22.1
M12 $\times$ 1.75	0.84	21.3	1.02	25.7
M14 $\times$ 2	0.96	24.3	1.16	29.3
M16 $\times$ 2	1.08	27.3	1.28	32.3
M18 $\times$ 2.5	1.20	30.3	1.45	36.6
M20 $\times$ 2.5	1.31	33.3	1.56	39.6
M22 $\times$ 2.5	1.43	36.3	1.68	42.6
M24 $\times$ 3	1.55	39.3	1.85	46.8

Rounding: round full thread depth and extra drill depth upward to $.01''$ or $.1 \text{ mm}$, then add for drill depth. Metric inch and millimeter columns are calculated independently; use one unit system consistently. Thread sizes and formula constants retain their exact values.

Inch UNC Threads Number 8 to 1 Inch

All depth values are in inches. Pitch = $1 \div$ threads per inch.

Thread size	Full thread depth (in)	Extra drill depth (in)	Tap Drill Depth (in)
#8-32	0.38	0.13	0.51
#10-24	0.41	0.13	0.54
#12-24	0.45	0.13	0.58
1/4-20	0.50	0.13	0.63
5/16-18	0.60	0.14	0.74
3/8-16	0.69	0.16	0.85
7/16-14	0.79	0.18	0.97
1/2-13	0.88	0.20	1.08
9/16-12	0.97	0.21	1.18
5/8-11	1.07	0.23	1.30
3/4-10	1.25	0.25	1.50
7/8-9	1.44	0.28	1.72
1-8	1.63	0.32	1.95

Application Notes

Engagement: the thread-depth formulas are proposed modeling defaults, not strength ratings. Verify the screw penetration, entrance chamfer, tolerance stack and required engagement for the actual screw and tapped material. Extra thread depth only adds engagement when the screw reaches it.

If shorter tap drills or flat bottom tap drills are required, call them out on our mfg prints.

Drill point: these depths end at the shoulder. The drill point extends farther; include it when checking remaining wall thickness and nearby passages. Fine threads require recalculation using their actual pitch.

Formulas are based on Co-op Tool Mfg Preferences.

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